A Comparative Analysis of Sparse Autoencoder and Activation Difference in Language Model Steering

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arXiv:2510.01246v1 [cs.CL] 24 Sep 2025

Abstract

Sparse autoencoders (SAEs) have recently emerged as a powerful tool for language model steering. Prior work has explored top-k SAE latents for steering, but we observe that many dimensions among the top-k latents capture non-semantic features such as punctuation rather than semantic attributes like instructions. To address this, we propose focusing on a single, most relevant SAE latent (top-1), eliminating redundant features. We further identify a limitation in constant SAE steering, which often produces degenerate outputs such as repetitive single words. To mitigate this, we introduce a token-wise decaying steering strategy, enabling more faithful comparisons with mean activation difference baselines. Empirically, we show that steering an SAE latent associated with reasoning reliably elicits step-by-step mathematical reasoning and enhances inference quality, functionally resembling the effect of appending a guiding token. Our results demonstrate that SAEs outperform mean activation difference methods on mathematical reasoning benchmarks and match their performance on IF-Eval.

1 Introduction

Steering language models by directly manipulating internal activations is an increasingly popular method for controlling model behavior without fine-tuning or prompt engineering (Stolfo et al., 2024; Postmus and Abreu, 2024) . Methods such as mean activation difference (**MeanActDiff**) (Stolfo et al., 2024) and sparse autoencoder (**SAE**) (Makelov, 2024; Mayne et al., 2024; O'Brien et al., 2024)-based steering have shown initial success in tasks including sentiment control (Tigges et al., 2024) and style transfer (Turner et al., 2023; Liu et al., 2024b; Scalena et al., 2024; von Rütte et al., 2024). Recently, some works have extended the scenarios to complex math reasoning and instruction-following tasks (Galichin et al., 2025; Wu et al., 2025b), demonstrating the robustness and versatility of SAE-based steering.

However, MeanActDiff lacks interpretability, offering little insight into the manipulated semantic features (Liu et al., 2024b). SAE-based steering provides a more interpretable alternative by exposing sparse, human-understandable activation features (Joseph Bloom and Chanin, 2024). Existing SAE steering methods often introduce noise and redundancy by selecting top-activated features without verifying their relevance (Shu et al., 2025; Rajamanoharan et al., 2024). In addition, most existing methods apply a constant steering strength throughout generation, which risks destabilizing generation or causing unnatural repetition, particularly in multi-step reasoning tasks, as shown in Figure 1. These limitations have prevented steering methods from being broadly adopted in more challenging scenarios, such as chain-of-thought math reasoning (Wei et al., 2022) and fine-grained instruction (Zhou et al., 2023) adherence, where precise and stable control is essential.

To address these challenges, we first propose an SAE feature selection mechanism based on subtracting top-activated feature sets, which isolates the dimensions most responsible for behavioral shifts while filtering out irrelevant activations. Second, we design a token-wise decaying steering strategy that dynamically reduces intervention strength during generation, improving stability and mitigating oversteering. We evaluate these methods on Gemma-2-2b and Gemma-2-9b for chain-of-thought math reasoning (GSM8K (Cobbe et al., 2021), SVAMP (Patel et al., 2021), MAWPS (Koncel-Kedziorski et al., 2016), ASDIV (Miao et al., 2020)) and on Gemma-2-9b-it for instruction following (IFEval (Zhou et al., 2023), covering JSON formatting, case sensitivity, multilingual responses, and lexical constraints). Our results show that decaying SAE steering im-

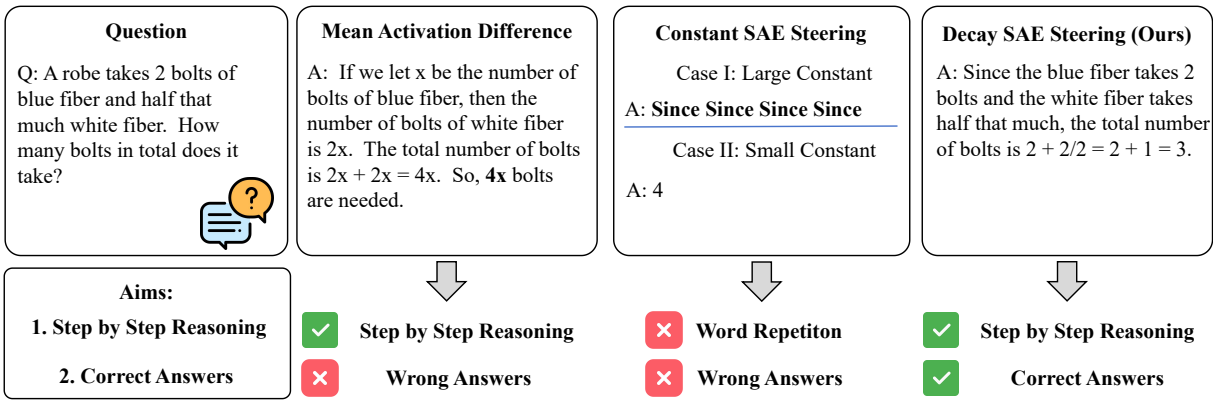

Figure 1: Mean Activation Difference (MeanActDiff) produces incorrect answers in math reasoning cases, while constant SAE steering often leads to word repetitions or wrong outputs. Our proposed decaying strategy applies the SAE steering vector progressively, yielding both step-by-step reasoning and correct answers, and enabling a fair comparison with MeanActDiff.

proves stability and achieves finer control compared to both MeanActDiff and constant SAE steering. In math reasoning, SAE steering reliably elicits step-by-step solutions without requiring few-shot prompts, outperforming MeanActDiff. In instruction following, our feature selection method more clearly identifies redundant latents, while in math reasoning further refinement may be needed. Finally, we observe that SAE steering behaves similarly to appending an additional guiding token to the prompt.

2 Related Works

Language Model Steering via Activation Editing Recent research into mechanistic interpretability explores direct control over language model outputs by intervening in their internal workings (Conmy et al., 2023; Rai et al., 2024). This is achieved by adding or subtracting steering vectors from hidden activations at specific layers to influence the style, tone, or content of the generated output (Subramani et al., 2022; Chalnev et al., 2024). A primary method for identifying these directions is the Mean Activation Difference (MeanActDiff) (Stolfo et al., 2024), which computes the average difference in activations between input pairs that elicit contrasting behaviors (e.g., generic vs. instruction-following responses) (Liu et al., 2024a). Several variants such as MeanActDiff-PCA and Linear Artificial Tomography (LAT) have been developed (Zou et al., 2025; Wu et al., 2025b).

Sparse Autoencoders for Intepretable Steering To overcome the lack of interpretability in MeanActDiff's global steering vectors, Sparse Autoencoders (SAEs) offer a more expressive alternative by learning a compressed and disentangled representation of a model's hidden activations (Lieberum et al., 2024; Gao et al., 2024). SAE training has followed diverse directions. Some works like Gated, Top-K and Switch SAE are trying to improve the SAE architecture (Gao et al., 2024; Rajamanoharan et al., 2024; Mudide et al., 2024), while others like LayerGroup, and Formal Language SAE (Ghilardi et al., 2024; Menon et al., 2024) are trying to improve the training strategies. These pre-trained SAEs (Lieberum et al., 2024; He et al., 2024; Chaudhary and Geiger, 2024) have enabled applications such as steering semantic instructions (Wu et al., 2025a), biological features (Farrell et al., 2024), reveal the hallucinations in LLMs (Ferrando et al., 2024), and study cross-lingual toxicity behaviours (Gallifant et al., 2025).

3 Methods

3.1 Preliminaries and Notation

Let $\mathcal{T} = \{\mathcal{T}_i\}_{i=0}^{K-1}$ denote a test set comprising K instances. Correspondingly, let $\mathcal{T}^* = \{\mathcal{T}_i^*\}_{i=0}^{K-1}$ be an augmented version of this set, where each instance is modified to elicit a desired behavior. In the context of mathematical reasoning, this augmentation is primarily achieved through in-context learning (ICL), where few-shot chain-of-thought examples are prepended to the input. For instruction following cases, we appended the instructions to the input as well. Let $\mathcal{F}$ represent a pretrained large language model. We define $\mathcal{A}_l(\cdot)$ as an operator that extracts the hidden activation vector (e.g., from an MLP or attention layer) from layer l of $\mathcal{F}$

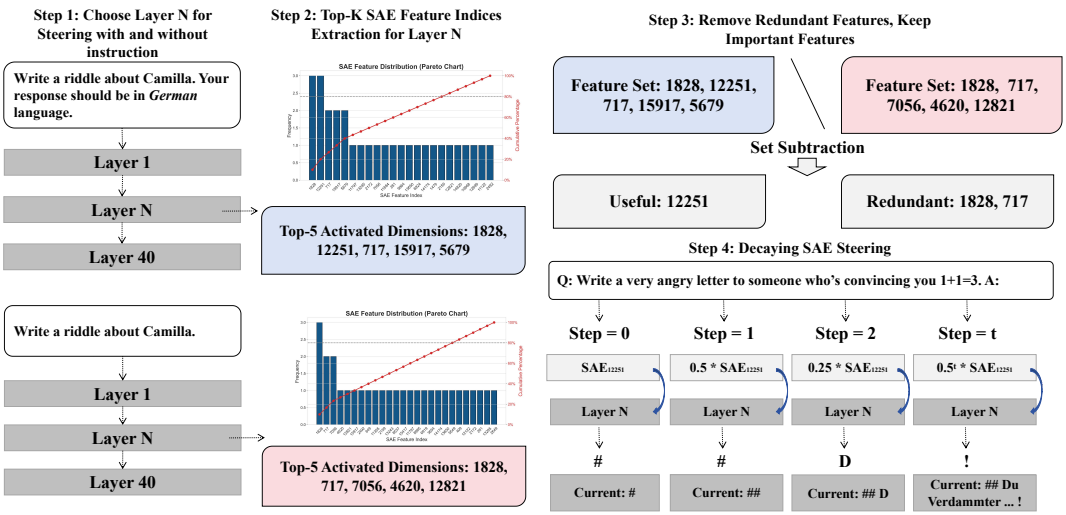

Figure 2: Pipeline of SAE feature extraction and token-wise decaying steering applied to question answering in **German**.

for a given input. Furthermore, let $\{\mathcal{S}_l\}$ be a set of pretrained Sparse Autoencoders, where each $\mathcal{S}_l$ is trained on the activations from layer l of $\mathcal{F}$. Each SAE contains a dictionary of interpretable features, $\{\mathbf{d}_{l,j}\}_{j=1}^{N_l}$, where $\mathbf{d}_{l,j}$ is the j-th feature vector (dictionary atom) for layer l and N_l is the total number of learned features. With this formalism, we can define the two primary steering methods.

Mean Activation Difference (MeanActDiff) Vector The MeanActDiff method generates a global steering vector by averaging the activation differences between paired examples from $\mathcal{T}^*$ and $\mathcal{T}$. For a given layer l, the MeanActDiff steering vector, $\mathbf{v}_l$, is formally defined as:

$$\mathbf{v}_l = \frac{1}{K} \sum_{i=0}^{K-1} \left(\mathcal{A}_l(\mathcal{F}(\mathcal{T}_i^*)) - \mathcal{A}_l(\mathcal{F}(\mathcal{T}_i)) \right) \quad (1)$$

This vector represents the average direction in activation space that corresponds to the behavior induced by the instructions or examples in $\mathcal{T}^*$.

Sparse Autoencoder (SAE) Features In contrast to deriving a vector from behavioral differences, a Sparse Autoencoder (SAE) discovers an overcomplete basis of interpretable features directly from the distribution of a model's activations. For a given layer l, an SAE is an encoder-decoder network trained to reconstruct the activation vector $\mathbf{h}_l = \mathcal{A}_l(\mathcal{F}(\mathcal{T}_i^*))$. The encoder maps the activation $\mathbf{h}_l$ to a high-dimensional, sparse latent vector $\mathbf{z}_l \in \mathbb{R}^{d_{sae}}$ (where $d_{sae} \gg d_{model}$):

$$\mathbf{z}_l = \mathrm{ReLU}(\mathbf{W}_e \mathbf{h}_l + \mathbf{b}_e) \quad (2)$$

where $\mathbf{W}_e \in \mathbb{R}^{d_{sae} \times d_{model}}$ is the encoder weight matrix and $\mathbf{b}_e$ is the encoder bias. The decoder then reconstructs the original activation from the sparse representation $\mathbf{z}_l$:

$$\hat{\mathbf{h}}_l = \mathbf{W}_d \mathbf{z}_l + \mathbf{b}_d \quad (3)$$

where $\mathbf{W}_d \in \mathbb{R}^{d_{model} \times d_{sae}}$ is the decoder weight matrix and $\mathbf{b}_d$ is the decoder bias. Crucially for steering, the columns of the trained decoder matrix $\mathbf{W}_d$ form an overcomplete dictionary of feature vectors, $\{\mathbf{d}_{l,j}\}_{j=1}^{d_{sae}}$. Each vector $\mathbf{d}_{l,j}$ represents a distinct, approximately monosemantic feature learned by the SAE. This provides a rich dictionary of potential steering vectors, from which one or more can be selected to guide generation.

3.2 Most Relevant SAE Feature Extraction

To isolate the SAE features causally linked to the behavior elicited by the augmented dataset $\mathcal{T}^*$, we propose a method to identify features that are differentially activated when the model processes $\mathcal{T}^*$ compared to the standard dataset $\mathcal{T}$. The objective is to find features that activate consistently for the augmented (e.g., chain-of-thought) inputs while remaining inactive for the standard inputs, thereby pinpointing the representations of the desired behavior. Our procedure, formalized in Algorithm 1, focuses on the model's state at the most critical juncture: the final token position of the input prompt. This pre-generation state is highly influential for the model's response.

Algorithm Description Algorithm 1 iterates through the paired instances from the paired